Matthew D. Woodson

matthew.d.woodson@gmail.com +1 586-703-6823

Data Engineering | Embedded Systems | ADAS | Robotics | Machine Learning | Artificial Intelligence | MLOps

AI/ML Software Engineer with 12+ years developing embedded systems all the way to agentic AI data pipelines. Demonstrated ability to lead engineering teams and deliver software and systems that directly addresses core user and stakeholder needs. Expertise spans MLOps, data warehousing, data processing pipelines, LLM/AI-agent powered data analysis, and evaluation systems.

Skills & Technologies

Languages: Python (Pandas, NumPy), C++, C, SQL, Cypher, Bash, MATLAB.
AI, ML & Data Engineering: LangChain/LangGraph, Agentic Workflows, Vision-Language Models, RAG, Neo4j, Google BigQuery, Apache Airflow & Superset, OpenCV, Streamlit
Cloud, DevOps & Infrastructure: Docker, Kubernetes, Jenkins, GitHub Actions, Ray, High-Performance Computing (HPC), Linux
Automotive & Systems Engineering: AUTOSAR, ISO 26262, MISRA, CAN, ROS, ADAS Perception Models, EKF Sensor Fusion, Occupancy Grids, Embedded Architectures.

Experience

Ford Motor Company

Senior AI/ML Engineer | *May 2025 – Present*

- Architected and currently deploying AI agents to triage BlueCruise autonomous driving events. Developed custom tools for the agent, including mapping tools, GIS queries, data extraction tools, and deterministic event analysis tools to increase accuracy.
- Develop agentic and LLM output evaluations to improve accuracy of data analysis and detect hallucinations
- Extensively evaluated Vision-language model such as Nvidia cosmos and Google gemma's performance in automotive applications.
- Engineered evaluation framework for the agent pipeline and developed detailed and helpful reporting formats and insights.
- Built, prompt engineered, and deployed an LLM-based warranty claim analysis tool that determines with high accuracy the category of a repair, the root cause, and the action that fixed the issue.
- Developed a Streamlit dashboard that allows users to drill into various categories, root causes, and actions taken across warranty claims.
- Sandboxed RAG-based Q&A system utilizing documents, warranty claims, semantic embedding, and vector search to help users gain new insight into warranty data.
- Developed a pilot Neo4j graph database to map complex relationships between vehicle build options, Diagnostic Trouble Codes (DTCs), and warranty claims.
- Engineered an LLM interface that converts natural language questions into Cypher/SQL queries, enabling management to identify part failures, repair procedures, and time-in-service trends.

- Maintain CI/CD pipeline and consistently improve AI tools in code development and review
- Developed data warehousing strategies and connected multi-modal models and data together

Senior Machine Learning Infrastructure Engineer | *Oct 2023 – May 2025*

- SAFe trained scrum lead and individual contributor for team of 7 people.
- Engineered data aggregation methods for training Birds Eye View surround perception models.
- Contributed to data warehousing, classification, and visibility of model training data utilizing technologies such as ROS, Google BigQuery, Apache Airflow, and Apache Superset.
- Architected end to end algorithm simulation environment to process vehicle logged data through BEV perception models and on downstream to upcoming ADAS safety features to hone precision and recall for those algorithms before test vehicles were ready.
- Co-designed an image degradation model training and evaluation pipeline to detect dirt or other camera obstructions including creating a new ontology of camera obstructions and reduced visibility environmental conditions.
- Scaled model evaluation pipelines utilizing Ray multiprocessing in High-Performance Computing (HPC) environments to handle large-scale concurrent data analysis.
- Debugged, trained, and evaluated image-based trailer detection neural network models

ADAS Algorithm Engineer | *April 2020 – Oct 2023*

- Productionized first-of-its-kind Pro Trailer Hitch Assist ADAS feature on multiple Ford vehicles.
- Owned and refined radar-based "Movable Occupancy Grid Map" and "Trailer Detector" C++ algorithms ensuring accuracy, requirements, and unit tests were consistently meeting high standards.
- Co-owned Extended Kalman Filter Sensor Fusion C++ algorithm, the core of Pro Trailer Hitch Assist.
- Conducted rigorous in-vehicle testing, data collection, bench testing, and ADAS module-level debugging running QNX operating system.
- Engineered a suite of vehicle data processing, algorithm output visualization, and GPS-based ground-truthing tools to speed up the vehicle testing to algorithm sign-off iterative loop.
- Explored ways to accelerate algorithm performance on ECU using DSPs and new libraries.

General Motors

Software Algorithm Engineer | *Sep 2017 – Nov 2019*

- Designed C and Matlab Simulink software for engine cooling components across vehicle lines and technologies
- Worked within AUTOSAR / ISO 26262 / MISRA standards
- Utilized INCA and VehicleSpy tools for in-vehicle software testing and debugging
- Executed HIL (hardware-in-the-loop) bench software testing of new software
- Created virtual unit tests for complete logic branching coverage
- Authored fan performance and temperature sensor diagnostics meeting OBD-II compliance
- Improved embedded software cybersecurity and robustness to loss of communication
- Reconstructed vehicle issues using data logs and custom plots to root cause issues
- Proposed and implemented project to convert MATLAB Simulink models to human maintained C code saving 30-50% of time needed for each change request
- Collaborated across multidisciplinary teams coordinating changes and knowledge sharing

FANUC America (Formerly FANUC Robotics)

Software Development Engineer Aug 2013 – Jan 2017

- Developed C software for embedded robot controller system
- Enhanced robot UI by coding multi-lingual features and new convenience menu options
- Performed root cause analysis and remediation to bug reports using register dumps, call stacks, and diagnostic logs
- Maintained software build and release server and scripts
- Completed Auto Backward Exit software feature for greatly enhanced robot error recovery
- Programmed robot motion, safety zones, and IO in multi-robot manufacturing cells
- Trained FANUC's iRVision system to locate parts for pickup or perform inspections

Education

Georgia Institute of Technology

Master of Science coursework: CS-7641 Machine Learning, CS-6200 Operating Systems

Kettering University

Bachelor of Science in Computer Engineering

Volunteer Activities

Ford Pride ERG – People Pillar Dec 2025 – Present

- Creating opportunities for connection globally for all regions of Ford Pride

Billiards Commissioner – Stonewall Sports Detroit Sep 2023 – Jan 2026

- Solo organizer and facilitator of a billiards league with 22 teams and 110 participants